

Compact $U(1)$ under refinement: an explicit total-variation bound in three dimensions, and why it fails in four

Result, 2026-09-27. For the heat-kernel (Villain) $U(1)$ lattice gauge theory on the periodic lattice $(\mathbb{Z}/N)^D$ of spacing a , physical side $L = Na$ and plaquette heat time t , the lattice measure splits exactly into a monopole-free part (the free photon together with the physical flux sectors of the torus) and a gas of lattice monopoles with Coulomb weights (Proposition 1; the duality of Banks, Myerson and Kogut). A bound on the lattice Laplacian then gives:

- $D = 3$ (**Theorem 2**). With $t = \lambda_3 a \leq 1$,

$$\|\mu_a - \mu_a^0\|_{\text{TV}} \leq \frac{2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}}{1 - e^{-\pi^2/(2\lambda_3 a)}},$$

where μ_a^0 is the monopole-free measure. The monopole free-energy density per physical volume lies between $a^{-3} \cdot 6e^{-2\pi^2/(3\lambda_3 a)}$ (up to a factor tending to one) and $a^{-3} \cdot 2e^{-\pi^2/(6\lambda_3 a)}/(1 - e^{-\pi^2/(2\lambda_3 a)})$. At fixed λ_3 and L everything compact vanishes faster than any power of a ; along trajectories with $\lambda_3 a \simeq c/\log(1/a)$ it stays finite. The flux sectors carry the weight $e^{-2\pi^2|m|^2/(\lambda_3 L)}$, independent of a .

- $D = 4$ (**Theorem 3, and a recorded failure**). With $t = g^2$, the same decomposition holds with flux-sector weight $e^{-2\pi^2|m|^2/g^2}$, independent of both a and L , and the monopole-loop free energy per lattice cell lies between $12e^{-\pi^2/g^2}$ and $8e^{-\pi^2/(8g^2)}/(1 - e^{-3\pi^2/(8g^2)})$, up to factors tending to one. The total-variation bound is then of order $(L/a)^4 e^{-\pi^2/(8g^2)}$, which grows as $a \rightarrow 0$ at fixed g . The criterion that closes the three-dimensional case fails here for an exact reason: the loop density per lattice cell is fixed by g alone. The four-dimensional continuum statement (Driver's renormalized free field) needs observables and a coupling renormalization, which a partition-function bound cannot supply.

This is the first step of the $D = 3$ programme of STATE carried out at the level of the measure, with explicit a , L and coupling, for the smallest compact group. It sharpens Theorem 5 of the series/parallel note (one step, rate $e^{-\pi^2/(8t)}$) into a bound on the whole lattice measure with rate $e^{-\pi^2/(6t)}$, and it puts numbers on the error-budget row of the zero-spacing note. The duality is established (Banks, Myerson and Kogut 1977, metadata) and the continuum limit at fixed coupling is Gross's theorem (Gross 1983, abstract as indexed); the explicit total-variation bound and the flux-sector bookkeeping are elementary consequences, with no novelty claimed.

1. The exact decomposition

Let E , P and C_3 be the links, plaquettes and 3-cells of the periodic hypercubic lattice $(\mathbb{Z}/N)^D$, d the coboundary on cochains, and $\Delta = dd^\top + d^\top d$ the Hodge Laplacian. The heat-kernel measure on link angles $\theta \in [0, 2\pi)^E$ is

$$\mu_a(d\theta) = \frac{1}{Z} \prod_{p \in P} k_t((d\theta)_p) \frac{d\theta}{(2\pi)^{|E|}}, \quad k_t(\phi) = \sum_{n \in \mathbb{Z}} e^{-tn^2/2} e^{in\phi} = \sqrt{\frac{2\pi}{t}} \sum_{n \in \mathbb{Z}} e^{-(\phi + 2\pi n)^2/(2t)}.$$

Write $\mathbb{R}^P = \text{im } d \oplus \text{im } d^\top \oplus \mathcal{H}$ (exact, coexact and harmonic 2-cochains; $\mathcal{H}$ is spanned by the cochains constant on each of the $\binom{D}{2}$ plaquette orientations). For $n \in \mathbb{Z}^P$ let $q = dn \in \mathbb{Z}^{C_3}$ (the monopole charges) and $n_{\mathcal{H}}$ the harmonic projection of n .

Proposition 1 (monopole decomposition). μ_a is a mixture

$$\mu_a = \frac{1}{Z} \sum_{[n] \in \mathbb{Z}^P/d\mathbb{Z}^E} w([n]) \nu_{[n]}, \quad w([n]) = \exp\left\{-\frac{2\pi^2}{t} \left(q^\top (dd^\top)^+ q + |n_{\mathcal{H}}|^2\right)\right\},$$

of probability measures $\nu_{[n]}$ on link angles, where $(dd^\top)^+$ is the pseudo-inverse on $\text{im } d$ and $|\cdot|$ the Euclidean norm on cochains. The classes $[n]$ with $q = 0$ make up the monopole-free measure μ_a^0 .

Proof. Insert the Poisson form of k_t and exchange sum and integral. The substitution $n \mapsto n + d\ell$, $\theta \mapsto \theta - 2\pi\ell$ ($\ell \in \mathbb{Z}^E$) groups the sum over n into classes and extends the angle integral to $\mathbb{R}^E$ modulo 2π times the integer closed 1-cochains, a lattice of full rank in the closed 1-cochains; the remaining integrand depends on θ only through $\omega = d\theta$. So the class $[n]$ contributes a constant times $\int_{\text{im } d} e^{-|\omega + 2\pi n|^2/(2t)} d\omega$. Decompose $n = n_{\text{ex}} + n_{\text{co}} + n_{\mathcal{H}}$ orthogonally. The exact part is absorbed by translating ω , and $|n_{\text{co}}|^2 = (dn)^\top (dd^\top)^+ (dn)$ because $n_{\text{co}} = d^\top (dd^\top)^+ dn$. The remaining Gaussian integral is the same for every class. $\square$

The classes are labelled by $q \in d\mathbb{Z}^P$ together with an integer flux vector $m \in \mathbb{Z}^{\binom{D}{2}}$ through the coordinate 2-tori (the integral cohomology of the torus is free). For fixed q the harmonic part is $n_{\mathcal{H}} = h(m) + y(q)$ with $h(m)$ the constant cochain of flux m and $y(q)$ an offset fixed by q . A cochain of flux $m_{\mu\nu}$ through each $(\mu\nu)$ torus has value $m_{\mu\nu}/N^2$ on each of the N^D plaquettes of that orientation, so

$$|h(m)|^2 = N^{D-4} \sum_{\mu < \nu} m_{\mu\nu}^2.$$

Two elementary facts. (i) Each Fourier mode of Δ on the hypercubic lattice has eigenvalue $\sum_{\mu} 4 \sin^2(k_{\mu}/2) \leq 4D$, so $\|dd^\top\| \leq 4D$ and $q^\top (dd^\top)^+ q \geq |q|^2/(4D)$ on $\text{im } d$. (ii) For y fixed, the theta-type sum $\sum_m e^{-\alpha|h(m)+y|^2}$ lies between $e^{-\alpha|y|^2} \sum_m e^{-\alpha|h(m)|^2}$ (pair m with $-m$) and $\sum_m e^{-\alpha|h(m)|^2}$ (its Fourier coefficients are positive).

The elementary monopole pair or loop. For a single plaquette p , the charge $q = d e_p$ is a nearest monopole pair ($D = 3$) or an elementary monopole loop ($D = 4$), and $q^\top (dd^\top)^+ q = |P_{\text{co}} e_p|^2$. By translation invariance each plaquette has the same projections, so $|P_{\text{ex}} e_p|^2 = \text{rank } d_1/|P|$ and $|P_{\mathcal{H}} e_p|^2 = \binom{D}{2}/|P|$. With $\text{rank } d_1 = |E| - (N^D - 1) - D$:

$$D = 3: |P_{\text{co}} e_p|^2 = \frac{1}{3} \left(1 - \frac{1}{N^3}\right), \quad D = 4: |P_{\text{co}} e_p|^2 = \frac{1}{2} \left(1 - \frac{1}{N^4}\right).$$

For $D = 3$ this agrees with the lattice Green's function: $G(0) - G(e_1) = \frac{1}{6}(1 - N^{-3})$, so a nearest pair has energy $2(G(0) - G(e_1))$.

2. Three dimensions: the theorem

Theorem 2. Let $D = 3$, $N = L/a$, $t = \lambda_3 a \leq 1$, and $R = Z/Z^0$ the ratio of the full partition function to its monopole-free part. Then

$$1 + 6N^3 \exp\left\{-\frac{2\pi^2}{3\lambda_3 a} - \frac{2\pi^2 a^2}{\lambda_3 L^3}\right\} \leq R \leq \exp\left\{\frac{2N^3 e^{-\pi^2/(6\lambda_3 a)}}{1 - e^{-\pi^2/(2\lambda_3 a)}}\right\},$$

and

$$\|\mu_a - \mu_a^0\|_{\text{TV}} \leq 1 - \frac{1}{R} \leq \frac{2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}}{1 - e^{-\pi^2/(2\lambda_3 a)}}.$$

The monopole-free part is the free photon on the torus with the flux sectors weighted by $\exp\{-2\pi^2|m|^2/(\lambda_3 L)\}$.

Proof. The flux weight is $|h(m)|^2 = N^{-1}|m|^2$ and $tN = \lambda_3 L$. Upper bound: by fact (ii) the sum over m at fixed q is at most its value at $q = 0$, so $R \leq \sum_{q \in d\mathbb{Z}^P} e^{-(2\pi^2/t)q^\top (dd^\top)^+ q}$. By fact (i) with $4D = 12$ each term is at most $e^{-(\pi^2/6t)|q|^2}$. Enlarge the sum to all of $\mathbb{Z}^{C_3}$, $|C_3| = N^3$, and use $\sum_{k \in \mathbb{Z}} e^{-\alpha k^2} \leq 1 + 2e^{-\alpha}/(1 - e^{-3\alpha})$ (from $k^2 - 1 \geq 3(k - 1)$) with $\alpha = \pi^2/(6t)$, then $\log(1 + x) \leq x$. Lower bound: keep $q = 0$ and the $6N^3$ nearest pairs $q = \pm d e_p$ ($3N^3$ plaquettes, two signs). Each pair has Coulomb energy at most $\frac{1}{3}$ and harmonic offset $|y|^2 = |P_{\mathcal{H}} e_p|^2 = N^{-3}$,

and fact (ii) gives the stated factor. Total variation: μ_a is the mixture $R^{-1}\mu_a^0 + (1 - R^{-1})\mu_a^{\text{rest}}$, so $\|\mu_a - \mu_a^0\|_{\text{TV}} \leq 1 - R^{-1} \leq \log R$. $\square$

Reading. In a box of fixed physical side, the lattice measure is within total variation $O((L/a)^3 e^{-\pi^2/(6\lambda_3 a)})$ of the monopole-free measure. At fixed λ_3 this tends to zero faster than any power of a . By the summable-error criterion of the refinement note, Proposition 4, the continuum question for the compact theory then reduces to that for the free photon with flux sectors, which is Gaussian; Gross's theorem is the continuum statement. The monopole free-energy density $L^{-3} \log R$ lies between the two bounds in the summary. It stays finite exactly when $e^{-c/(\lambda_3 a)} \sim a^3$, that is along $\lambda_3 a \simeq c/(3 \log(1/a))$ with c between $\pi^2/6$ and $2\pi^2/3$: the trajectories on which a Debye mass survives (Göpfert–Mack 1982, abstract as indexed). The exponents are bounds; the monopole self-energy of the infinite lattice, $2\pi^2 G(0) \approx 4.99$ with $G(0) \approx 0.2527$, lies between them.

Corollary 2' (iteration reduces to the free field). Let $a_n = 2^{-n}a_0$ and let p_{n0} be the composite of the $3n$ directional halvings from spacing a_n to a_0 (multiplying fine links). Then

$$\|p_{n0*}\mu_{a_n} - p_{n0*}\mu_{a_n}^0\|_{\text{TV}} \leq \frac{2(L/a_n)^3 e^{-\pi^2/(6\lambda_3 a_n)}}{1 - e^{-\pi^2/(2\lambda_3 a_n)}},$$

and $p_{n0*}\mu_{a_n}^0$ is the blocked free photon with the flux sectors of the torus, which blocking maps to the same sectors. So stability under iteration, the open clause of Hypothesis P(α) in the series/parallel note, is reduced for $U(1)$ in $D = 3$ to exact Gaussian blocking: the compact part never has to be iterated.

Proof. A pushforward does not increase total variation, and Theorem 2 bounds the distance before the pushforward. The flux of a coarse 2-torus is the sum of the fluxes of the fine plaquettes it contains, so the sector label m is preserved. $\square$

The Gaussian comparison that remains, between the blocked free photon and the free photon at spacing a_0 , is not small in total variation: the two quadratic forms differ at order one on modes of the lattice scale, and agree to relative order $a^2 K^2$ on smooth modes (Proposition 2 and bound (6) of the series/parallel note). It must be made through smooth observables, where it is explicit because both sides are Gaussian.

3. Four dimensions: the same bound, and why it fails

Theorem 3. Let $D = 4$, $N = L/a$, $t = g^2$ with $g^2 \leq 1$. Then

$$1 + 12N^4 \exp\left\{-\frac{\pi^2}{g^2} - \frac{2\pi^2}{g^2 N^4}\right\} \leq R \leq \exp\left\{\frac{8N^4 e^{-\pi^2/(8g^2)}}{1 - e^{-3\pi^2/(8g^2)}}\right\},$$

$\|\mu_a - \mu_a^0\|_{\text{TV}} \leq \log R$, and the flux sectors carry the weight $\exp\{-(2\pi^2/g^2) \sum_{\mu < \nu} m_{\mu\nu}^2\}$.

Proof. As for Theorem 2, with $4D = 16$, $|C_3| = 4N^4$, $|h(m)|^2 = |m|^2$, $6N^4$ plaquettes and two signs for the elementary loops, loop energy at most $\frac{1}{2}$, and harmonic offset N^{-4} . $\square$

The recorded failure. In four dimensions the heat time is the coupling itself, so at fixed g the monopole-loop free energy per lattice cell is a fixed number between the two bounds, and the total-variation bound grows like $(L/a)^4$. The three-dimensional route, closeness of the whole lattice measure to its Gaussian part, is unavailable in four dimensions for an exact reason, and the free-energy density it controls is a vacuum constant. The known continuum statement is different in kind: the compact theory converges on its current sector to a renormalized free electromagnetic field (Driver 1987, abstract as indexed), with the monopole loops renormalizing the charge. A four-dimensional theorem in the insertion language therefore has to control observables and a coupling shift per step, as Hypothesis P(α) does, rather than the measure. This is the four-dimensional row of the error budget in the zero-spacing note, now with the density per cell made explicit: between $12e^{-\pi^2/g^2}$ and $8e^{-\pi^2/(8g^2)}$ up to factors tending to one.

4. Consequence for STATE

Item 1 of STATE gains a measure-level theorem for $U(1)$ in $D = 3$ with explicit a , L , λ_3 (Theorem 2), and a recorded failure of the same route in $D = 4$ with its reason (Theorem 3). The non-abelian step that corresponds to Theorem 2 needs a replacement for the Poisson decomposition: for $SU(2)$ the exact kernel (11) of the series/parallel note has image terms of both signs, so the mixture of Proposition 1 is not available and the Laplace-remainder route of its Proposition 7 remains the path.